***This docoment captures my thinking process and flow.***

***Thoughts:***

„Lepšie mobilné rozhranie pre Emu na ceste automaticky znamená lepšie dáta pre Filipa v kancelárii.“

co ak ema bude chciet vyplnat digitalne ale jozef jej povie ze to nechce robit na svojom mobile ?

aky je rozdiel v rieseni situacii ked ema sposobila buracku alebo ked jozef sposobil buracku, ako to ovplyvni emu v kazdej situaciii? komu vyplaca v kazdom pripade poistovna?

Po vyplneni celeho formulara sa moze eme poslat na mail pdf – ako sa to vsak posle jozefovi ?

Co vsetko viem spravit po nehode neskor(doma) ?

# DISCOVER: research and As-Is situation analysis

## 1. User Mindset & Environment

When a car accident happens (even a minor parking lot bump or a simple intersection collision), the user is far from a normal state:

- Physiological and psychological state**:** Adrenaline rush, shock, shaking hands, anger, fear of financial consequences or confrontation with the other driver.
- Context of use: Outdoors on the road, in rain, darkness, cold, or direct sunlight (bad screen visibility), traffic noise, time pressure (honking cars).

## 2. Current Process Steps (As-Is User Journey)

- **Impact and initial shock:** Stopping the vehicle, turning on hazard lights, checking for injuries, putting on a reflective vest, and placing the warning triangle. The immediate thought: 'Do I have to call the police?'
- **Confrontation and document search :** Talking to the other driver, searching the glove compartment for a printed EAS form (which can be missing or lacking a pen), and looking for the green card / mandatory insurance and vehicle registration.
- **Chaotic documentation:** Drivers take bad photos (often just a close-up of a scratch, missing overall context of the intersection, license plates, - making photos insufficient for the insurer later).
- **Filling out the paper EAS form:** Filling out a two-page carbon copy form on a car hood (Vehicle A in the blue column, Vehicle B in the yellow column). Checking 17 complex legal circumstances and trying to sketch the intersection and driving trajectory by hand.

- **Signing and departure**: Separating carbon copies, feeling uncertain whether filling it out unintentionally admitted fault they didn't mean to accept.
- **Insurance report and black hole (Days to weeks post-accident):** Realizing at home that the paper form still needs to be reported to the insurer via web or phone, re-entering all data, or personally handing out the document and waiting weeks in uncertainty without knowing the claim status.

## 3. Pain Points

### Critical Severity

- **Decision Paralysis in First Minutes:** Drivers in shock do not know what to do first for their safety and when calling the police is legally mandatory (e.g., injuries, fluid leaks, disagreement on fault, public property damage).
- **High Cognitive Load of Paper EAS Form:** The form looks like a tax return. It requires rewriting data the insurer already has (policy number, VIN, vehicle owner data) or data that could be read from a license plate or QR code.
- **Incomplete and Poor Evidence:** Once cars are moved, accident site photos cannot be retaken. Drivers don't know how to photograph properly (missing 4 angles, license plates, road context), leading to claim delays or rejections.
- **Synchronous Paper Filling by Strangers:** Two stressed drivers crowd over a single piece of paper. If one makes a mistake, crossing it out creates distrust and unnecessary delay in a dangerous spot (road shoulder).

### High Severity

- **Accident sketching (field 13):** Asking a stressed person to hand-draw an intersection layout, traffic signs, and vehicle vectors on their knee is a UX disaster. Sketches are often incomprehensible to adjusters.
- **Misleading terminology (field 12):** The distinction between 'entering a roundabout' and 'driving on a roundabout', or 'turning right' vs. 'changing lanes' confuses drivers, who fear signing acknowledges legal fault.
- **Post-accident 'black hole' :** After submitting the report, the user has no idea what is happening, whether the insurer needs anything else, when the inspector arrives, or when they can take their car to a repair shop.

**Edge cases**

- private property / parked car: record details if owner present; mandatory police call if owner missing (hit and run prevention).
- guardrails / animals: state/road administrator property (nds/ssc); mandatory police call for public property damage; claim handled via casco insurance, not pzP.
- chain collision (3+ vehicles): messy paper forms; police usually called due to unclear initial impact sequence.

- ux solution (triage screen): 3 visual cards on start:
    1. vehicle collision (standard 2-car flow + add vehicle option).
    2. object/fence/guardrail/animal impact (checks for public property -> triggers 158 police call button + single car mode).
    3. chain crash (recommends police).

**Web app or native app (pwa)**

- prefered: mobile-first progressive web app (pwa).
- fast access in seconds:
    1. qr code on digital green card in apple/google wallet.
    2. sms link from insurer helpline or google search.
- advantages: loads instantly in browser (safari/chrome), no device storage needed, accessible for drivers with different insurers.

**onboarding & pre-filling**

- automatic user data retrieval from insurance database (name, id, address, license plate, vin, policy number).
- fast authentication under stress:
    - method 2: enter license plate + phone number -> 4-digit sms otp code -> profile & policy loaded
    - method 1: qr code ?

G**ps location & timestamp**

- html5 geolocation api in modern browsers (safari/chrome).
- user taps "confirm location" -> permission prompt -> reverse geocoding fetches exact street/km mark and system timestamp automatically -> user verifies pin on map.

**voice input and converting into text**

- issue: speech-to-text fails near noisy roads and heavy traffic; editing garbled text causes frustration.
- Possible solutions:
    1. tap-to-build (zero typing): smart chips for quick thumb taps:
        - location: [intersection] [parking lot] [straight road] [roundabout]
        - my car: [stopped] [moving straight] [turning] [reversing]
        - other car: [failed to yield] [rear-ended] [changed lanes]

2. audio attachment only: hold microphone button for voice note without text transcription on-site -> raw audio saved for claims adjuster, optional ai summary reviewed later safely at home.

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# DEFINE : User personas (in the file directory), Problem statement

**Problem statement:**

**For the Driver (Ema):**
*After a collision, drivers are overwhelmed by shock and stress, making rational decision-making and clear thinking extremely difficult. They do not know the correct protocol or what priorities to follow—such as whether calling the police is legally mandatory, what evidence must be captured immediately on-site, and what administrative details can wait until they are safe at home. Faced with a complex paper form on a busy road, they struggle to hand-draw traffic sketches and accurately document the incident.*

**For the Insurer (Filip):**
*Adjusters regularly receive paper reports that are either incomplete, incorrectly filled out, or completely illegible. Hand-drawn collision sketches are frequently incomprehensible, and attached photo documentation is often insufficient, failing to capture critical vehicle angles and road context. Consequently, adjusters waste excessive time manually transcribing data and resolving disputes instead of settling claims efficiently. Also Processing car accident claims relies heavily on manual work that could easily be automated.*

## How might we change things

**less typing and automated collection:**

- The app already knows your car and profile from insurer login.
- No need to re-type the other driver's data: either scan their license plate + driver's license/green card via camera OCR, or generate a QR code/link for them to open on their phone (without downloading an app, via Web Clip / Instant App) to sync data in 10 seconds.

**Step-by-Step crisis assistant - guidance:**

- The app first verifies safety (3 quick Yes/No questions: Anyone injured? Blocking traffic?) and immediately fetches exact location and time via GPS (eliminating Fields 1 and 2 of EAS).

### Guided evidence capture:

- Instead of an abstract 'Add Photos' button, the camera opens with visual overlays: 'Step 1: Photograph the full scene from 5 meters', 'Step 2: Photograph the impact area', 'Step 3: Photograph the other car's license plate.'
- Instead of typing long descriptions with shaking hands, use Voice-to-Text: 'Describe in your own words what happened, AI structures the report and selects the correct circumstances.'

### Asynchronous Flow (On-scene vs. Post-scene split):

- Key service design insight: On the road under stress, we only capture critical data (GPS, time, site photos, identification, and confirmation—taking 10 minutes cca). All administrative details (repair shop selection, bank account, detailed notes) can be completed in the comfort of home 2 hours later, with a live status tracker in the app (like Uber or package tracking).

**Design Principles**

1. **Calming design:** reduce panic, lower stress, showing only one simple decision or question per screen to prevent mental overload (one thing at a time), simple colour palette, simple/plain language – easily understandable, safety guidance – give clear feedback right away on what steps to take first, removing the fear of doing something wrong.

2. **As leat typing as possible:** Eliminate the need to type long text, Automatic location & time – fetch the exact street address, *Pre-filled policy data:* Automatically pull policyholder info, vehicle model, and insurance numbers from existing database records*, interactive tags* to quickly set the scene without opening a keyboard. *Camera scanning:* Allow instant scanning of driver's licenses and registration cards using camera instead of typing names and IDs manually.

3. **On-scene vs. Post-scene split:**

    - **Phase 1** – Critical data (Fast & Essential):
    - Confirm safety and emergency status (police necessity check).
    - Log GPS coordinates and timestamp.
    - Take necessary scene photos before moving cars.
    - Quickly link with the other driver via QR code or quick link.
    - **Phase 2** – post-scene admin (At home):
    - Fill in secondary details later (bank account / IBAN, preferred repair shop, detailed personal notes).
    - Review the compiled summary with a clear head before final submission.

4. **Guided visual reconstruction & evidence capture***: Intuitive Map-Based Scene Builder* (Eliminate the frustration of hand-drawing intersections on paper.

*(*Drivers can clearly tap damaged car panels (bumpers, doors, lights) simultaneously )

*Step-by-Step Camera Guidance* (guided photo taking)

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# DEVELOP (Ideation, Wireframing & UX Architecture): Mapping Out Collaboration, Interactions, and Core Screens

**ON A MAP**

**(Ema vs. Jozef)**

**1. parallel work without blocking –** ema sends a QR code to Jozef and they can both work parallely

- **ema (host)** confirms details, taps damaged parts on car a (2d blueprint), takes photos
- **jozef (guest)** scans license/plate on his phone, taps damaged parts on car b, takes photos
- **shared map:** ema creates the initial scene layout as host, placing both car a and car b
- **jozef** receives ema's draft and either he will confirm her layout or make his own if he disagrees

**2. three possible scenarios**

- **scenario a: ideal state**
    - ema creates map -> jozef reviews on phone -> taps [ i agree ] -> fast, zero friction
- **scenario b: minor adjustment**
    - ema places car in wrong lane -> jozef says so -> ema adjusts position or jozef updates -> taps [ i agree ]
- **scenario c: conflict (dispute mode)**
    - jozef taps [ disagree / submit my version ] -> moves vehicles to his version -> system saves both versions (version a & version b) without blocking submission
    - warning displayed: statements differ: both versions saved for adjuster. check photos or call police (158)

**4. approvement fot claim adjuster**

- replaces messy paper sketches and notes (
- adjuster sees verified gps, photos, and clear outcomes:
    - match -> 1 shared map with mutually confirmed stamp

- mismatch -> side-by-side comparison of version a vs. version b, easily verified via objective photo evidence

**The  companion concept – the GUY**

Throughout the entire process, minimalist illustration of a friendly assistant appears at the top of the screen He says what shoul be done or how. To lower stress and explain why that specific step matters.

**COMPLETE SCREEN-BY-SCREEN FLOW (9 Core Screens) –** generated with AI (mine would be too chaotic)

**PHASE 0: Entry Point & Authentication**

**Screen 1: Login & Client Portal Home**

- **What happens:** Ema is standing by the roadside and opens her insurer's mobile web portal (or taps a saved wallet link). She logs in quickly via an SMS one-time code / FaceID (or enters her license plate + phone number so she doesn't have to recall a password under stress).
- **Screen Layout:**
  - Right at the top (above all regular policies and invoices), there is a prominent, high-contrast **Emergency Action Card**: 🚨 Had an accident? Report a Claim Now.
  - Directly below it, her insured vehicle is pre-selected (Škoda Octavia • BL-123AB) alongside a secondary quick-action button: Need a tow truck? Call Assistance.
  - Standard portal items (My Policies, Payments, Documents) are pushed to the bottom and visually muted so they don't distract her during an emergency.
- **User Action:** Ema taps the large **[ Report an Accident ]** button.

**PHASE 1: Combined Calming Safety & Incident Triage**

**Screen 2: Safety Reminder & Incident Triage (Combined Screen)**

- **What happens:** Tapping the report button switches the interface into a clean, distraction-free Focus Mode. By combining the safety checklist with the incident type selector on a single screen, a panicked first-time driver (Ema) gets immediate life-saving guidance, while an experienced driver who has already secured the scene isn't slowed down by mandatory checkboxes and can proceed in 2 seconds with a single tap.
- **Screen Layout:**

- **1. Top – Calming Companion:** A subtle assistant illustration with a gentle breathing animation, saying in a reassuring speech bubble: *"Take a deep breath, Ema. Make sure you're safe first, then tap what happened. This will only take 5 minutes."*
- **2. Upper-Middle – Non-Blocking Safety Reminder Banner:**
  - A compact visual card displaying 3 scannable safety icons (no mandatory ticking required—acts as an instant visual check):
    - 🦺 **Vest & Hazards:** *Hazards on & reflective vest worn*
    - 📐 **Triangle:** *Placed 50m–100m behind car*
    - 🤝 **Calm Approach:** *No injuries & ready to connect*
  - Embedded emergency action inside the banner: **[ 🚨 Injuries or danger? Call 112 / 158 ]**.
- **3. Lower-Middle – Incident Type Selector (What kind of accident occurred?):**
  - Large selectable cards that advance the flow:
    - **Card 1:** [ 🚙💥🚙 Collision with another vehicle ] *(Standard two-vehicle flow – proceeds directly to QR pairing)*.
    - **Card 2:** [ 🚙💥🌲 Hit an object, fence, or parked car ] *(Triggers an inline sub-question: "Is the property owner present?" If no, or if public infrastructure like guardrails/signs is damaged, prompts a mandatory 158 Police call)*.
    - **Card 3:** [ 🚙🚙🚙 Chain collision (3+ vehicles) ] *(Prompts police assistance recommendation)*.
- **Primary CTA: [ Continue ]**

**PHASE 2: Driver Pairing & Automated Data (Pairing & Auto-Data)**

**Screen 3: Connect with the Other Driver via QR Code / Link**

- **What happens:** Ema links her session with the second driver (Jozef) so both statements merge under a single Claim ID, allowing each driver to fill out their vehicle details in parallel on their own phone.
- **Screen Layout:**
  - **Companion Character:** *"Show this screen to the other driver. Once they scan the code, you can both fill out your parts at the same time—no app download needed!"*
  - **Center Visual:** Large **QR Code** + **[ Share Link via SMS ]** button.

- **Live Status Pill:** ⏳ Waiting for the other driver to connect... -> updates upon scan to: 🟢 Driver B (Jozef) connected!
- **Bottom Fallback Option:** *"Other driver doesn't have a smartphone or prefers to use yours?"* -> Secondary Button: **[ Fill it out on one phone together ]** (unlocks camera OCR scanning for Jozef's driver's license and license plate on Ema's phone).
- **Primary CTA: [ Start Filling Details ]**

**Screen 4: Automated Location, Timestamp & Vehicle Verification**

- **What happens:** Instead of manually filling out Fields 1–8 of the paper EAS form, HTML5 GPS location, exact time, and Ema's vehicle details are pulled automatically from her phone and insurer profile.
- **Screen Layout:**
  - **Companion Character:** *"I've automatically filled in your car details, exact time, and GPS location so you don't have to type."*
  - **Card 1 – Location & Time (Auto-GPS):** Mini map preview with a pin, exact street address (Bajkalská 29, Bratislava), and timestamp (Today, 14:32) + a small Edit location link in case the vehicles moved slightly off the road.
  - **Card 2 – Your Vehicle (Pre-filled from portal):**
    - Driver: Ema Kováčová (with a toggle for *"Someone else was driving"*).
    - Vehicle: Škoda Octavia • BL-123AB • Policy #884920.
  - **Primary CTA: [ Confirm My Details ]**

**PHASE 3: Visual Evidence & Scene Reconstruction (Damage, Photos & Map)**

**Screen 5: Mark Damaged Areas – 2D Unfolded Vehicle Blueprint (Replacing EAS Field 10 & 11)**

- **What happens:** Ema taps the specific panels on her car that were damaged during the impact (while Jozef simultaneously does the same for his car on his phone). For smaller or specific parts not shown on the main diagram (e.g., side mirror, cracked headlight only, windshield wiper), she can quickly select a chip or type it into an optional field below.
- **Screen Layout:**
  - **Companion Character:** *"Tap on the parts of YOUR car that were hit or scratched. No need to rotate anything—all sides are right here."*
  - **Center Visual: 2D Unfolded Vehicle Blueprint** (top-down roof in the center, surrounded by unfolded left/right doors and fenders, front bumper and hood at the top, trunk and rear bumper at the bottom).

- Tapping Front Bumper & Grill and Front Lights & Hood highlights those sections in high-contrast coral red.
- **Specific / Minor Part Input (Below Blueprint):**
  - Label: Specific part damaged? (e.g., side mirror, headlight only, wheel rim)
  - Quick chips: [ Left Wing Mirror ] [ Right Wing Mirror ] [ Headlight Only ] [ Wheel / Tire ]
  - Optional text input field: [ 🖍️ Specify other damaged part... ]
- **Selection Counter above button:** 2 damaged areas selected.
- **Primary CTA: [ Continue to Guided Photos ]**

**Screen 6: Guided Camera Evidence Capture**

- **What happens:** The web app accesses the smartphone camera and guides Ema through 4 required shots, ensuring claims adjuster Filip receives complete, verifiable photo documentation.
- **Screen Layout:**
  - **Companion Character (in top camera bar):** *"Follow the outlines on the screen. Good wide photos prevent any disputes later!"*
  - **Camera Viewfinder:** Features a semi-transparent wireframe overlay in the center with a clear prompt: Step 1 of 4: Step back 5 meters and take a wide photo showing both cars and the road.
  - **Bottom Required Photo Slots (Checklist):**
    1. Wide Scene View (Both cars & road)
    2. Point of Contact (Where cars touched)
    3. Your Damage: Front Bumper & Hood *(dynamically generated from her selections in Screen 5!)*
    4. Other Vehicle's License Plate
  - **Primary CTA (once all 4 shots are taken): [ Save Photos & Continue ]**

**Screen 7: Interactive Map Scene Reconstruction & Remarks (Replacing EAS Field 12, 13 & 14)**

- **What happens:** Instead of sketching lanes and arrows by hand on paper, Ema (as the session host) drags two vehicle tokens onto the real GPS map, selects their movement states, and adds any personal remarks.
- **Screen Layout:**

- **Companion Character:** *"Almost done! Drag the two cars onto the map to show where they met, then tap how they were moving."*
- **Interactive Map Window (GPS Street Layout):** Top-down satellite/vector view of the exact road or intersection. Two distinct vehicle tokens with visible headlight beams (to clearly indicate the front of each car) are placed on the map: **Blue Car A (Mine)** and **Red Car B (Other)**. Users can drag and rotate them.
- **Movement Selector below Map (Two side-by-side columns):**
  - Left Column CAR A (MINE): [ ⬆️ Moving Forward ] [ 🛑 Stopped ] [ ⬇️ Reversing ]
  - Right Column CAR B (OTHER): [ ⬆️ Moving Forward ] [ 🛑 Stopped ] [ ⬇️ Reversing ]
- **Context Smart Chips (Replacing the 17 legalistic checkboxes - Field 12):**
  - [ Intersection ] [ Roundabout ] [ Parking Lot ] [ Changing Lanes ] [ Yield Sign ]
- **My Remarks Section (Replacing EAS Field 14):**
  - Label: Additional remarks or observations (Optional):
  - Quick chips: [ I acknowledge fault ] [ Dashcam video available ]
  - Input field: [ 🖍️ Write a short remark or 🎙️ record voice note... ]
- **Primary CTA: [ Send Scene to Jozef for Review ]**

**PHASE 4: Mutual Review & Post-Scene Transition**

**Screen 8: Mutual Review & On-Scene Confirmation**

- **What happens:** Jozef receives Ema's map layout on his phone and taps **[ I Agree ]** (or taps *Submit my version* in case of disagreement, which saves both versions without causing a roadside deadlock). Both drivers now view the combined on-scene summary (including what the other driver filled out and their remarks) and confirm it digitally to close the roadside part. *(Note: Strict legal identity/signature verification is documented under project limitations/future scope).*
- **Screen Layout:**
  - **Companion Character:** *"Great teamwork! Both of you have reviewed the scene. Confirm below, and you're free to leave the road safely."*
  - **Status Banner:** ✅ Jozef confirmed the scene reconstruction. *(Optional UI note showing the Dispute fallback state if statements differ).*
  - **Shared Evidence Summary Card (Full Transparency for Both Drivers):**
    - 📍 Location & Time verified (Bajkalská 29, 14:32)

        - 🟦 Vehicle A (Ema): Front Bumper, Hood • Remarks: "Had right of way"
        - 🟧 Vehicle B (Jozef): Rear Bumper • Remarks: "None"
        - 📸 8 photos & GPS map reconstruction saved
    - **Calming Legal Reassurance Note:** ℹ️ Confirming this report locks the factual evidence from the scene and sends a copy to both phones via SMS. Liability is assessed later by the insurance company.
    - **Confirmation Action: [ ⭕ Confirm & Finish On-Scene Part ]**

**Screen 9: Home Claim Tracker & Post-Scene Admin Completion**

- **What happens:** Two hours later, Ema is safe and calm at home. She opens her portal (or clicks the SMS magic link) to finish non-urgent administrative details and track her claim in real time—eliminating the post-accident "Black Hole".
- **Screen Layout:**
    - **Companion Character:** *"Welcome home, Ema! Now that you're safe, just pick your preferred repair shop and bank account so Filip from claims can process your payout."*
    - **Live Claim Status Tracker (Delivery-style progress stepper):**
        1. 🟢 **On-Scene Evidence Captured** *(Completed at 14:38)*
        2. 🔵 **Home Details & Service Selection** *(In Progress – Action Needed)*
        3. ⚪ **Adjuster Review (Assigned to Filip)** *(Estimated within 24 hours)*
        4. ⚪ **Repair & Payout Approved**
    - **Three Quick Post-Scene Task Cards:**
        - 🔧 Choose Authorized Repair Shop (Select a partner garage from a map)
        - 💳 Payout Bank Account (IBAN) (Pre-filled from profile—1 tap to confirm)
        - 🎙️ Optional Voice or Text Note (Add extra context in a quiet environment at home)
    - **Primary CTA: [ Submit Final Claim Details ]**

**Summary of All 9 Core Screens:**

1. **Portal Home:** High-visibility *Report an Accident* emergency card at the very top.
2. **Combined Safety & Incident Triage:** Companion calms Ema down + non-blocking 3-icon safety reminder + scenario selector (*2 vehicles* vs. *object/fence/guardrail* vs. *3+ chain crash*).
3. **QR Pairing:** Connects Ema and Jozef under one shared Claim ID.

4. **Auto-Location & Car:** Pre-filled GPS, timestamp, and policy vehicle data.
5. **2D Damage Blueprint + Specific Part Input:** Tap-to-select exterior car panels on an unfolded 2D diagram + quick chips/text field for specific details like side mirrors or lights.
6. **Guided Camera:** Step-by-step viewfinder overlays for required photo angles.
7. **Map Scene Builder & Remarks:** Drag-and-drop 2 cars on a GPS map + *Forward / Stopped / Reversing* buttons + Smart Chips + Remarks field.
8. **Mutual Review & Confirmation:** Transparent summary of both drivers' inputs, Propose & Review confirmation with Jozef, and safe departure from the road.
9. **Home Claim Tracker:** Asynchronous completion of IBAN/repair shop at home + live claim status tracking.

**DELIVER: figma, html**